

CSCC10H3: Human-Computer Interaction (HCI)

Assignment 2: Heuristic Usability Evaluation

Due Date: 11:59 pm Sunday, July 7, 2024

Worth: 10% of your final grade.

Overview:

Heuristic usability evaluation entails finding usability problems in a user interface design. It involves having a small set of evaluators examine the interface and judge its compliance with recognized usability principles (i.e., the heuristics).

In this assignment, you will play a role of an expert evaluator for the Toronto Pearson Airport website (<https://www.torontopearson.com/en>)

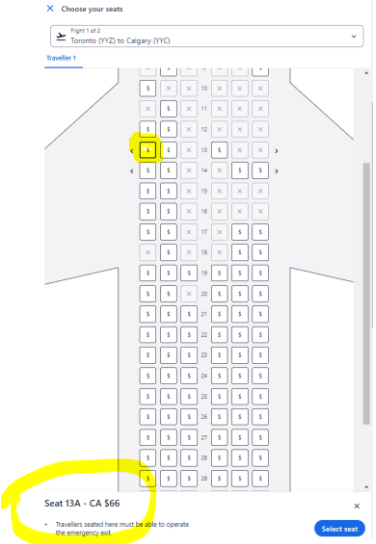
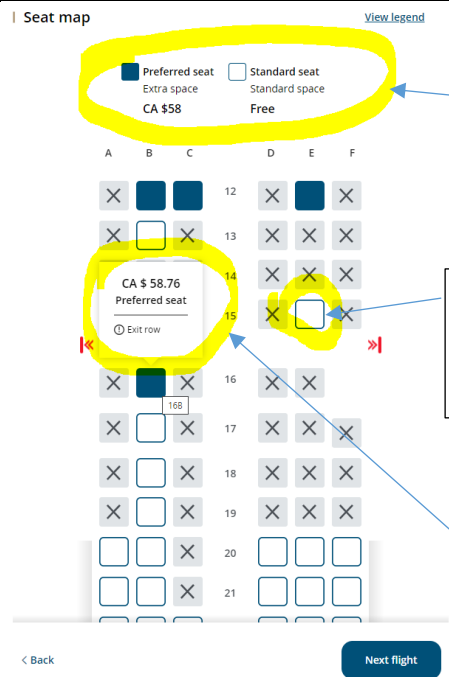
You will initially work alone as an expert evaluator and then work in groups of three to combine and report your findings. As an expert evaluator, you will need to examine the user interface using Nielsen's ten usability principles outlined at: <https://www.nngroup.com/articles/ten-usability-heuristics/> and also listed in Appendix A.

Part 1: Individual evaluation

Evaluate the Toronto Pearson website independently, identifying as many usability problems as you can and the violated heuristics for each problem. You should not discuss the evaluation results with others to avoid bias. Take good notes as you will need to share them with your group and attach it as Appendices to the final report.

Part 2: Discussion and Reporting

Meet with your group to discuss your findings. Combine all three sets of usability problems into **one** overall list (no duplicates). Here is a sample to illustrate one problem on a travel website (www.expedia.ca):

Usability Problem	On Expedia.ca, when selecting a seat, the user has to click on the \$ box to see the price show up at the bottom of the screen (i.e., information is not visible easily). Also, it's not possible for the user to see if any seats are lower in price or free unless they click on each \$ box.
Heuristic(s) Violated	H6
Screenshot	
Here is a seat selection screenshot from the AirCanada.com site which addresses some of the violations mentioned above.	 Legend, so the user knows what the blue box and empty box means. Empty box representing seats are free, and it states that when the user hovers over the box. When the user hovers over the blue box, they see the price along with whether it's an exit row, bulkhead row, etc.

Together come up with at least five improvements based on the problems identified and indicate which heuristics it would satisfy (it can satisfy multiple heuristics).

What to submit:

A written report describing your evaluations. The report should include the following:

1. Title page should include your group members' full names and Student IDs.
2. A description of as many usability problems with a screenshot (you may list multiple problems under one screenshot) and indicate which heuristics are violated (e.g., H1, H2, etc.). You may find a problem that violates multiple heuristics.
3. List and describe **at least** five of the problems identified above and suggest improvements. Indicate which heuristic(s) will the improvement satisfy (e.g., H1, H2, etc.) and why. Include screenshots/sketches of the interface elements to explain your recommendations.
4. Appendices should include each evaluator's raw notes from the independent heuristic evaluations – Part 1. Identification of problems, descriptions, screenshots/sketches, etc. This can be hand-written and scanned, or digitally generated.

Note: All group members will receive identical marks for this assignment, which will be calculated based on the final report, including the Appendices.

Submission Instructions:

Please submit a single PDF file on Quercus with the following naming convention: “A3.[last names of your group members].pdf (for example, “A3.Nizam-Smith-MacDonald.pdf”). Only one member of the team should submit the assignment on Quercus.

Grading Rubric:

Components	Marks Distribution
Overall Presentation of Report and Clarity	15
Usability Problems & Details	35
Five Problems with Suggested Improvements	35
Appendices	15
Total Marks	100

Appendix A: Usability Heuristics (Nielsen, 1994)

<https://www.nngroup.com/articles/ten-usability-heuristics/>

1. Visibility of system status (H1):

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

2. Match between system and the real world (H2):

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.

3. User control and freedom (H3):

Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.

4. Consistency and standards (H4):

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

5. Error prevention (H5):

Even better than good error messages is a careful design which prevents a problem from occurring in the first place.

6. Recognition rather than recall (H6):

Make objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

7. Flexibility and efficiency of use (H7):

Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8. Aesthetic and minimalist design (H8):

Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9. Help users recognize, diagnose, and recover from errors (H9):

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10. Help and documentation (H10):

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.